THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 150	Find the new time first, then the angle: each hour mark is 30°.
2	**(a)** 1170	$360n + 90$.
3	**(a)** 180	Net turn.
4	**(a)** 30	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
5	**(a)** 25	The minute hand turns 6° per minute.
6	**(c)** 4	Each line gives one straight angle.
7	**(d)** 72	$360^\circ \div 5$.
8	**(d)** 150	Multiply the angle between neighbours by the number of gaps.
9	**(b)** 90	Find the new time first, then the angle: each hour mark is 30°.
10	**(a)** 1890	$360n + 90$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hour hand points to 4 on the dial; smaller angle $= 120^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
12	Total $= 360 \times 2 + 90 = 810^\circ$ (1); Right angles $= (810) \div 90 = 9$ (1).	1+1
13	Each pair of rays makes one angle (1); $\frac{8 \times 7}{2} = 28$ (1).	1+1
14	Hour hand points to 6 on the dial; smaller angle $= 180^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand points to 1 on the dial; smaller angle $= 30^\circ$ (1); Larger angle $= 360 - 30 = 330^\circ$ (1); Reflex angle: the larger angle 330° (1).	1+1+1
16	Minutes $= 90 \div 6 = 15$ (1); Right angles $= 90 \div 90 = 1$ (1); Time shown: 2:15 (1).	1+1+1
17	Slices $= 2 \times 2 = 4$ (1); Angle $= 360 \div 4 = 90^\circ$ (1); Type: right (1).	1+1+1
	OR: Hour hand points to 5 on the dial; smaller angle $= 150^\circ$ (1); Larger angle $= 360 - 150 = 210^\circ$ (1); Reflex angle: the larger angle 210° (1).	1+1+1
18	Minutes $= 180 \div 6 = 30$ (1); Right angles $= 180 \div 90 = 2$ (1); Time shown: 4:30 (1).	1+1+1
	OR: Slices $= 2 \times 3 = 6$ (1); Angle $= 360 \div 6 = 60^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Hour hand: 2 hour marks of $30^\circ = 60^\circ$ (1); (b) Minute hand has completed 2 full turns $= 720^\circ$ and stands at 12 (1); (c) Smaller angle $= 60^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 180 - 45 + 90 = 225^\circ$ (1); Direction 225° clockwise from North (1); To face North again: turn clockwise 135° more (1); Check: $225 + 135 = 360^\circ$ (2).	1+1+1+2
20	(a) Hour hand: 4 hour marks of $30^\circ = 120^\circ$ (1); (b) Minute hand has completed 4 full turns $= 1440^\circ$ and stands at 12 (1); (c) Smaller angle $= 120^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Net $= 270 - 90 + 135 = 315^\circ$ (1); Direction 315° clockwise from North (1); To face North again: turn clockwise 45° more (1); Check: $315 + 45 = 360^\circ$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 270	Net turn.
2	**(d)** 60	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
3	**(a)** 40	The minute hand turns 6° per minute.
4	**(a)** 5	Each line gives one straight angle.
5	**(b)** 60	$360^\circ \div 6$.
6	**(c)** 160	Multiply the angle between neighbours by the number of gaps.
7	**(d)** 30	Find the new time first, then the angle: each hour mark is 30°.
8	**(b)** 2610	$360n + 90$.
9	**(d)** 90	Net turn.
10	**(b)** 120	The hands are placed symmetrically about the 12-6 line, so the angle is the same.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Total $= 360 \times 3 + 180 = 1260^\circ$ (1); Right angles $= (1260) \div 90 = 14$ (1).	1+1
12	Each pair of rays makes one angle (1); $\frac{9 \times 8}{2} = 36$ (1).	1+1
13	Hour hand points to 7 on the dial; smaller angle $= 150^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
14	Total $= 360 \times 4 + 270 = 1710^\circ$ (1); Right angles $= (1710) \div 90 = 19$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand points to 7 on the dial; smaller angle $= 150^\circ$ (1); Larger angle $= 360 - 150 = 210^\circ$ (1); Reflex angle: the larger angle 210° (1).	1+1+1
16	Minutes $= 270 \div 6 = 45$ (1); Right angles $= 270 \div 90 = 3$ (1); Time shown: 5:45 (1).	1+1+1
17	Slices $= 2 \times 4 = 8$ (1); Angle $= 360 \div 8 = 45^\circ$ (1); Type: acute (1).	1+1+1
	OR: Hour hand points to 8 on the dial; smaller angle $= 120^\circ$ (1); Larger angle $= 360 - 120 = 240^\circ$ (1); Reflex angle: the larger angle 240° (1).	1+1+1
18	Minutes $= 360 \div 6 = 60$ (1); Right angles $= 360 \div 90 = 4$ (1); Time shown: 8:00 (1).	1+1+1
	OR: Slices $= 2 \times 5 = 10$ (1); Angle $= 360 \div 10 = 36^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Hour hand: 5 hour marks of $30^\circ = 150^\circ$ (1); (b) Minute hand has completed 5 full turns $= 1800^\circ$ and stands at 12 (1); (c) Smaller angle $= 150^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Net $= 135 - 225 + 270 = 180^\circ$ (1); Direction 180° clockwise from North (1); To face North again: turn clockwise 180° more (1); Check: $180 + 180 = 360^\circ$ (2).	1+1+1+2
20	(a) Hour hand: 7 hour marks of $30^\circ = 210^\circ$ (1); (b) Minute hand has completed 7 full turns $= 2520^\circ$ and stands at 12 (1); (c) Smaller angle $= 150^\circ$ (1); (d) obtuse angle (2).	1+1+1+2
	OR: Net $= 90 - 180 + 225 = 135^\circ$ (1); Direction 135° clockwise from North (1); To face North again: turn clockwise 225° more (1); Check: $135 + 225 = 360^\circ$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Lines and Angles** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 55	The minute hand turns 6° per minute.
2	**(d)** 6	Each line gives one straight angle.
3	**(a)** 45	$360^\circ \div 8$.
4	**(a)** 135	Multiply the angle between neighbours by the number of gaps.
5	**(b)** 120	Find the new time first, then the angle: each hour mark is 30°.
6	**(b)** 3330	$360n + 90$.
7	**(a)** 135	Net turn.
8	**(a)** 150	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
9	**(d)** 75	The minute hand turns 6° per minute.
10	**(d)** 7	Each line gives one straight angle.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Each pair of rays makes one angle (1); $\frac{10\times9}{2} = 45$ (1).	1+1
12	Hour hand points to 9 on the dial; smaller angle $= 90^\circ$ (1); Hour hand turns $2 \times 30 = 60^\circ$ in 2 hours (1).	1+1
13	Total $= 360 \times 5 + 90 = 1890^\circ$ (1); Right angles $= (1890) \div 90 = 21$ (1).	1+1
14	Each pair of rays makes one angle (1); $\frac{12\times11}{2} = 66$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hour hand points to 10 on the dial; smaller angle $= 60^\circ$ (1); Larger angle $= 360 - 60 = 300^\circ$ (1); Reflex angle: the larger angle 300° (1).	1+1+1
16	Minutes $= 450 \div 6 = 75$ (1); Right angles $= 450 \div 90 = 5$ (1); Time shown: 9:15 (1).	1+1+1
17	Slices $= 2 \times 6 = 12$ (1); Angle $= 360 \div 12 = 30^\circ$ (1); Type: acute (1).	1+1+1
	OR: Hour hand points to 11 on the dial; smaller angle $= 30^\circ$ (1); Larger angle $= 360 - 30 = 330^\circ$ (1); Reflex angle: the larger angle 330° (1).	1+1+1
18	Minutes $= 540 \div 6 = 90$ (1); Right angles $= 540 \div 90 = 6$ (1); Time shown: 11:30 (1).	1+1+1
	OR: Slices $= 2 \times 9 = 18$ (1); Angle $= 360 \div 18 = 20^\circ$ (1); Type: acute (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Hour hand: 10 hour marks of $30^\circ = 300^\circ$ (1); (b) Minute hand has completed 10 full turns $= 3600^\circ$ and stands at 12 (1); (c) Smaller angle $= 60^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 225 - 90 + 135 = 270^\circ$ (1); Direction 270° clockwise from North (1); To face North again: turn clockwise 90° more (1); Check: $270 + 90 = 360^\circ$ (2).	1+1+1+2
20	(a) Hour hand: 11 hour marks of $30^\circ = 330^\circ$ (1); (b) Minute hand has completed 11 full turns $= 3960^\circ$ and stands at 12 (1); (c) Smaller angle $= 30^\circ$ (1); (d) acute angle (2).	1+1+1+2
	OR: Net $= 315 - 135 + 90 = 270^\circ$ (1); Direction 270° clockwise from North (1); To face North again: turn clockwise 90° more (1); Check: $270 + 90 = 360^\circ$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.